Resuelto 2: sistema $x \equiv 5 \pmod{11}$, $x \equiv 12 \pmod{13}$, $x \equiv 7 \pmod{17}$ I

$x \equiv 5 \mod 11$
$x \equiv 12 \mod 13$
$x \equiv 7 \mod 17$

2431

$m_1 = 221$ $\quad$ $221 \mod 11 = 1$

$m_2 = 187$ $\quad$ $187 \mod 13 = 5$

$m_3 = 143$ $\quad$ $143 \mod 17 = 7$

$1 \times y_1 \equiv 1 \mod 11$ $\quad$ $\text{extgcd}(1, 11)$ $\quad$ $y_1 = 1$ $\quad$ $1 = 11(0) + 1(1)$

$5 y_2 \equiv 1 \mod 13$ $\quad$ $\text{extgcd}(5, 13)$ $\quad$ $y_2 = 8$

$7 y_3 \equiv 1 \mod 17$ $\quad$ $5 \times 1 \mod 17$ $\quad$ $y_3 = 5$

$\text{extgcd}(5, 13) = 1$

$5 \mod 13 = 5$
$13 \mod 5 = 3$
$5 \mod 3 = 2$
$3 \mod 2 = 1$

$3 = 13(1) + 5(-2)$
$2 = 5(1) + 3(-1)$
$\boxed{1 = 3(1) + 2(-1)}$

$1 = (1)(13(1) + 5(-2)) + (5(1) + 3(-1))(-1)$

$1 = (1)(13(1) + 5(-2)) + (5(1) + (13(1) + 5(-2))(-1))(-1)$

$1 = 13(1+1) + 5(-2 - 1 - 2)$

$1 = 13(2) + 5(-5)$

$s = -5$

$x \equiv -5 \times 1 \mod 13$

$x \equiv -5 \mod 13 = 8$

$\text{ext}(7, 17)$

~~7 mod 17 = 7~~
$17 \mod 7 = 3$
$7 \mod 3 = 1$

$3 = 17 + 7(-2)$
$1 = 7 + 3(-2)$

$1 = 7 + (17 + 7(-2))(-2)$
$1 = 7(5) + 17(-2)$

$x = 5 \times 1 \mod 17$
$x \equiv 5 \mod 17 = 5$

$M_1 = 221$, $y_1 = 1$

$M_2 = 187$, $y_2 = 8$

$M_3 = 143$, $y_3 = 5$

$a_1 = 5$ $a_2 = 12$ $a_3 = 7$

$$X = (221 \times 1 \times 5 + 187 \times 8 \times 12 + 143 \times 5 \times 7) \bmod M$$

$2183 + K \times 2431$

$24062 \bmod M$

$24062 \bmod 2431 = 2183$

Resuelto 2: sistema $x \equiv 5 \pmod{11}$, $x \equiv 12 \pmod{13}$, $x \equiv 7 \pmod{17}$ I

$2183 \bmod 11 = 5$

$2183 \bmod 13 = 12$

$2183 \bmod 17 = 7$